

Feature Documentation — Merged 3-Class Model (v16)

grass / loose / paved

WalkSensePlace

Every feature is computed **per step** (one gait cycle, from one total-pressure peak to the next, $\sim 0.5\text{--}1.2\text{ s}$ at $f_s = 62.5\text{ Hz}$). Sensors used: 12 plantar pressure pads, a 3-axis accelerometer, and a 3-axis gyroscope (the magnetometer is dead and unused). Steps that span a surface change or a **transition/wait** tag are excluded, so each step lies on a single surface.

Two conventions appear throughout:

- **“norm” / session-normalised** — divided by a within-recording scale (e.g. that step’s mean total pressure) so the value doesn’t scale with body weight or how tightly the insole was worn.
- **amplitude-invariant** — the metric is a ratio or shape number, unaffected by overall size.

The model uses **48 features**. The F values are ANOVA class-separation scores on the full data (higher = separates grass/loose/paved better). Person-confound features that are computed but **dropped** are listed at the end.

Notation

For one step of length L samples, let p_t be total pressure; $s_{i,t}$ the value of pad i ($i = 1, \dots, 12$); heel sum $h_t = \sum_{i \in \mathcal{H}} s_{i,t}$ and forefoot sum $f_t = \sum_{i \in \mathcal{F}} s_{i,t}$ over the heel set $\mathcal{H}$ and forefoot set $\mathcal{F}$. Accelerometer axes are a_t^x, a_t^y, a_t^z with magnitude $\|a_t\| = \sqrt{(a_t^x)^2 + (a_t^y)^2 + (a_t^z)^2}$; gyroscope $g_t^{\{x,y,z\}}$ analogously. $\bar{x} = \frac{1}{L} \sum_t x_t$ is the mean. $\tilde{p}$ denotes the total-pressure curve resampled to a fixed length and divided by its mean (a unit-scale shape). The *flat-foot instant* is $t^* = \arg \min_t \|g_t\|$ (foot momentarily still), where the accelerometer points along gravity.

Several features are *step-to-step* quantities: for a per-step scalar $u^{(k)}$ (step index k within a recording+foot sequence), the rolling 3-step standard deviation is

$$\text{roll3_std}(u)^{(k)} = \sqrt{\frac{1}{3} \sum_{j=k-2}^k (u^{(j)} - \bar{u}^{(k)})^2}, \quad \bar{u}^{(k)} = \frac{1}{3} \sum_{j=k-2}^k u^{(j)}. \quad (1)$$

1 Foot-tilt variability (the strongest features)

From a ZUPT-gated orientation estimate: at the flat-foot instant t^* the foot tilt is read from the accelerometer on two orthogonal axes,

$$\theta_A = \text{atan2}(a_{t^*}^x, \sqrt{(a_{t^*}^y)^2 + (a_{t^*}^z)^2}), \quad \theta_B = \text{atan2}(a_{t^*}^z, \sqrt{(a_{t^*}^x)^2 + (a_{t^*}^y)^2}). \quad (2)$$

- **orient_rest_a_roll3_std** ($F \approx 1612$) and **orient_rest_b_roll3_std** ($F \approx 889$) — the **step-to-step variability** of resting foot tilt, i.e. $\text{roll3_std}(\theta_A)$ and $\text{roll3_std}(\theta_B)$ via Eq. (??). High values mean the foot lands at a different orientation each step — the signature of uneven/compliant ground. The single most class-differentiable features, and they generalise across

people. (The *absolute* tilt θ_A, θ_B is a per-person mount offset and is *not* a feature — only its step-to-step variability is kept.)

- **orient_range_a / orient_range_b** — the range of foot tilt *within a single stance*, $\max_t \theta_{A,t} - \min_t \theta_{A,t}$: how much the foot rocks while planted.

2 Accelerometer texture / impact (vibration and hardness)

Let x be a step-length signal, detrended, with real-FFT power spectrum $P_k = |X_k|^2$ at frequencies f_k ($k = 1, \dots, K$). Define the normalised spectrum $\rho_k = P_k / \sum_j P_j$.

$$\text{spectral centroid} = \frac{\sum_k f_k P_k}{\sum_k P_k}, \quad \text{HF fraction} = \frac{\sum_{k: f_k \geq 5 \text{ Hz}} P_k}{\sum_k P_k}, \quad (3)$$

$$\text{spectral entropy} = \frac{-\sum_k \rho_k \ln \rho_k}{\ln K}, \quad \text{jerk ratio} = \frac{\text{RMS}(\Delta x)}{\text{RMS}(x - \bar{x})} = \frac{\sqrt{\frac{1}{L-1} \sum_t (x_t - x_{t-1})^2}}{\sqrt{\frac{1}{L} \sum_t (x_t - \bar{x})^2}}. \quad (4)$$

- **accel_jerk_ratio** ($F \approx 1205$) — jerk ratio of the vertical axis a^y (amplitude-invariant); sharp/jerky strikes on hard ground, damped on compliant.
- **accel_hf_frac** ($F \approx 549$) — HF fraction of $\|a_t\|$; higher on firm surfaces.
- **accel_spec_centroid** ($F \approx 517$) — spectral centroid of $\|a_t\|$; shifts up with harder impacts.
- **accel_spec_entropy** ($F \approx 492$) — spectral entropy of $\|a_t\|$ (tonal vs broadband texture).
- **accel_zcr** — zero-crossing rate of a^y , $\frac{1}{L-1} \sum_t \mathbf{1}[\text{sign}(a_t^y - \bar{a}^y) \neq \text{sign}(a_{t-1}^y - \bar{a}^y)]$.
- **accel_mag_strike** — peak $\|a_t\|$ in the impact region $\max_{t \leq L/3} \|a_t\|$.

3 Gyroscope motion (foot rotation & stability)

- **gyro_x_std** ($F \approx 1046$) — standard deviation of the x angular rate over the step, $\sigma_{g^x} = \sqrt{\frac{1}{L} \sum_t (g_t^x - \bar{g}^x)^2}$: side-to-side foot rotation/wobble.
- **gyro_x_strike / gyro_y_strike / gyro_z_strike** — angular rate at the strike instant (rotational “kick” at contact).
- **gyro_y_std / gyro_z_std** — angular-rate std on the other two axes.
- **gyro_hf_frac** ($F \approx 397$) — HF fraction (same formula) of $\|g_t\|$.
- **gyro_mag_strike** — $\|g_t\|$ at the strike instant.

4 Stance timing & pressure-curve shape

On the unit-scale total curve $\tilde{p}$ (session-normalised shape):

- **stance_duration** ($F \approx 838$) — loaded time per step, $\#\{t : p_t > \frac{1}{2}\bar{p}\}$; compliant ground softens/lengthens contact.

- **loading_rate / loading_rate_norm** — peak rise rate $\max_t \Delta p_t$ (raw and /scale); hard loads faster.
- **rise_slope / fall_slope** — pressure rise and fall slopes.
- **slope_asym** — asymmetry between loading and unloading slopes.
- **time_to_peak_frac** — $\arg \max_t p_t / L$ (timing of peak loading).
- **gu_dip_depth** — mid-stance dip depth of $\tilde{p}$, $1 - \frac{\min_{t \in \text{mid}} \tilde{p}_t}{\max_t \tilde{p}_t}$; deeper on grass.
- **gu_load_first_frac** — $\sum_{t \leq L/2} \tilde{p}_t / \sum_t \tilde{p}_t$.
- **gu_push_peak** — height of the push-off (second) peak of $\tilde{p}$.
- **gu_load_slope / gu_unload_slope** — $\max_t \Delta \tilde{p}_t$ (loading) and $\min_t \Delta \tilde{p}_t$ (unloading).

5 Weight distribution (heel vs forefoot)

- **heel_fore_ratio** ($F \approx 372$) — $\bar{h} / \bar{f}$ (front-to-back load).
- **heel_impulse_norm** ($F \approx 394$) — $\sum_t h_t / \text{scale}$ (time-under-load at heel).
- **heel_skew / heel_kurt** — skewness and kurtosis of the heel series,

$$\text{skew} = \frac{\frac{1}{L} \sum_t (h_t - \bar{h})^3}{\sigma_h^3}, \quad \text{kurt} = \frac{\frac{1}{L} \sum_t (h_t - \bar{h})^4}{\sigma_h^4}.$$

- **heel_ripple_frac** — high-frequency “ripple” fraction of the heel signal.
- **midstance_diff** — $h_{t^*} - f_{t^*}$ at the flat-foot instant.
- **gu_fore_early_frac** — forefoot loading share in early stance, $\sum_{t \leq t_0} \tilde{f}_t / \sum_t \tilde{f}_t$.
- **gu_fore_heel_toff** — $(\arg \max_t \tilde{f}_t - \arg \max_t \tilde{h}_t) / L$.

6 Pressure-map distribution & dynamics (across the 12 pads)

Treat the 12 pads as a spatial map. Let v_i be a per-pad load (e.g. mean over stance, or the flat-foot value) and $q_i = v_i / \sum_j v_j$ its normalised share.

$$\text{spatial entropy} = \frac{-\sum_{i=1}^{12} q_i \ln q_i}{\ln 12}, \quad \text{participation} = \frac{1}{12} \cdot \frac{1}{\sum_{i=1}^{12} q_i^2}. \quad (5)$$

- **gu_spatial_entropy_stance** ($F \approx 510$) and **gu_spatial_entropy_ff** — normalised Shannon entropy of the pad distribution (whole stance / at flat-foot). High = load spread evenly; low = concentrated. Compliant ground conforms and spreads load.
- **gu_participation_ff** — normalised inverse participation ratio (effective fraction of pads bearing load at flat-foot).

- **gu_active_sensors** — $\#\{i : v_i > 0.1 \max_j v_j\}$ at flat-foot (contact breadth).
- **gu_cop_wander** — mean frame-to-frame change of the normalised map, $\frac{1}{L-1} \sum_t \sum_i |Q_{i,t} - Q_{i,t-1}|$ with $Q_{\cdot,t}$ the per-frame normalised map: how much load shifts around the sole.
- **gu_press_jitter** — mean per-pad temporal jitter, $\text{mean}_i \text{std}_t(\Delta^2 s_{i,t}) / \text{scale}$.
- **stance_duration_roll3_std / loading_rate_roll3_std / impact_mag_roll3_std** — step-to-step variability (Eq. (??)) of stance duration, loading rate, and impact.

Dropped (computed but excluded from the model)

Removed because cross-user permutation testing showed they act as a **person fingerprint** and *hurt* generalisation to a new person:

- **Per-sensor pressure ratios** — `pressure_01_mean_ratio ... pressure_12_mean_ratio`, plus `heel_mean_ratio`, `fore_mean_ratio`, `heel_pmax_ratio`, `fore_pmax_ratio`.
- **Raw magnitudes** — `total_p_mean`, `total_pmax`, `impact_mag`, `impact_mag_norm`, `heel_impulse` (scale with body weight / donning tightness).

How to read the model, in one line

The features that carry the surface signal — and survive to a new person — are **step-to-step foot-tilt variability**, **accelerometer texture/jerk**, **gyro rotation variability**, **stance timing**, and **pressure-map spread**. Together these describe *compliance* (soft vs firm) very well, which is why grass-vs-hard is reliable; they describe *fine firm-surface texture* weakly, which is why loose-vs-paved remains hard.